

```
import pandas as pd
df = pd.read_csv("sales_data.csv")
df
```

	OrderID	Month	City	Product	Category	Quantity	Price	Sales	CustomerID
0	1000	November	Mumbai	Laptop	Electronics	5	50000	250000	C016
1	1001	April	Pune	Refrigerator	Appliances	2	30000	60000	C044
2	1002	December	Mumbai	T-Shirt	Fashion	7	600	4200	C003
3	1003	January	Mumbai	Mobile Phone	Electronics	4	15000	60000	C033
4	1004	October	Mumbai	T-Shirt	Fashion	4	600	2400	C046
...	...	...	...	...	...	...	...	...	...
495	1495	October	Mumbai	Headphones	Electronics	5	2000	10000	C030
496	1496	November	Bangalore	Mobile Phone	Electronics	9	15000	135000	C030
497	1497	May	Pune	Laptop	Electronics	6	50000	300000	C011
498	1498	December	Bangalore	Refrigerator	Appliances	5	30000	150000	C046
499	1499	March	Mumbai	Refrigerator	Appliances	6	30000	180000	C019

500 rows × 9 columns

```
df.columns
```

```
Index(['OrderID', 'Month', 'City', 'Product', 'Category', 'Quantity', 'Price',
      'Sales', 'CustomerID'],
      dtype='object')
```

```
df.dtypes
```

	0
OrderID	int64
Month	object
City	object
Product	object
Category	object
Quantity	int64
Price	int64
Sales	int64
CustomerID	object

dtype: object

```
df.shape
```

```
(500, 9)
```

```
df.describe()
```



	OrderID	Quantity	Price	Sales
count	500.000000	500.000000	500.000000	500.000000
mean	1249.500000	5.720000	17229.600000	100118.800000
std	144.481833	2.748948	18298.593413	124346.227507
min	1000.000000	1.000000	600.000000	600.000000
25%	1124.750000	4.000000	2000.000000	6000.000000
50%	1249.500000	6.000000	15000.000000	30000.000000
75%	1374.250000	8.000000	30000.000000	150000.000000
max	1499.000000	10.000000	50000.000000	500000.000000

df[df["City"]== "Mumbai"]

	OrderID	Month	City	Product	Category	Quantity	Price	Sales	CustomerID
0	1000	November	Mumbai	Laptop	Electronics	5	50000	250000	C016
2	1002	December	Mumbai	T-Shirt	Fashion	7	600	4200	C003
3	1003	January	Mumbai	Mobile Phone	Electronics	4	15000	60000	C033
4	1004	October	Mumbai	T-Shirt	Fashion	4	600	2400	C046
6	1006	May	Mumbai	Mobile Phone	Electronics	7	15000	105000	C022
...	...	...	...	...	...	...	...	...	...
466	1466	December	Mumbai	T-Shirt	Fashion	8	600	4800	C017
470	1470	October	Mumbai	Mobile Phone	Electronics	1	15000	15000	C029
473	1473	December	Mumbai	Refrigerator	Appliances	3	30000	90000	C034
495	1495	October	Mumbai	Headphones	Electronics	5	2000	10000	C030
499	1499	March	Mumbai	Refrigerator	Appliances	6	30000	180000	C019

119 rows × 9 columns

df[df["Category"]== "Fashion"]

	OrderID	Month	City	Product	Category	Quantity	Price	Sales	CustomerID
2	1002	December	Mumbai	T-Shirt	Fashion	7	600	4200	C003
4	1004	October	Mumbai	T-Shirt	Fashion	4	600	2400	C046
11	1011	November	Delhi	T-Shirt	Fashion	4	600	2400	C046
16	1016	November	Mumbai	T-Shirt	Fashion	3	600	1800	C035
27	1027	February	Bangalore	Shoes	Fashion	10	2500	25000	C030
...	...	...	...	...	...	...	...	...	...
472	1472	April	Bangalore	T-Shirt	Fashion	10	600	6000	C031
481	1481	November	Bangalore	Shoes	Fashion	6	2500	15000	C050
483	1483	February	Delhi	Shoes	Fashion	5	2500	12500	C048
490	1490	July	Bangalore	Shoes	Fashion	4	2500	10000	C049
492	1492	February	Pune	T-Shirt	Fashion	1	600	600	C012

175 rows × 9 columns

df.max()



	0
OrderID	1499
Month	September
City	Pune
Product	T-Shirt
Category	Fashion
Quantity	10
Price	50000
Sales	500000
CustomerID	C050

dtype: object

df.min()

	0
OrderID	1000
Month	April
City	Bangalore
Product	Headphones
Category	Appliances
Quantity	1
Price	600
Sales	600
CustomerID	C001

dtype: object

df["Sales"].max()

500000

df.groupby("Month")["Sales"].sum().idxmax()

'May'

df.groupby("Month")["Sales"].sum().max()

6208100

df.groupby("Month")["Sales"].sum().idxmin()

'January'

df.groupby("Product")["Quantity"].sum().idxmax()

'T-Shirt'

df.groupby("Product")["Sales"].sum().idxmax()

'Laptop'

df.groupby("City")["Sales"].sum().idxmax()

'Bangalore'

df["City"].value_counts().idxmax()



```
'Bangalore'
```

```
df.groupby("OrderID")["Sales"].sum().mean()
```

```
np.float64(100118.8)
```

```
df.groupby("OrderID")["Sales"].sum().max()
```

```
500000
```

```
df.groupby("OrderID")["Sales"].sum().min()
```

```
600
```

```
df.groupby("CustomerID")["Sales"].sum().idxmax()
```

```
'C019'
```

```
df.groupby("Month")["Sales"].sum()
```

Sales	
Month	
April	3185100
August	3900100
December	3250200
February	4419900
January	2674800
July	3493200
June	4243900
March	4503700
May	6208100
November	5700600
October	3729200
September	4750600

```
dtype: int64
```

```
df.groupby("Category")["Sales"].sum().idxmax()
```

```
'Electronics'
```

```
df.groupby("City")["Sales"].mean()
```

Sales	
City	
Bangalore	112763.235294
Delhi	88381.060606
Mumbai	102326.890756
Pune	96286.725664

```
dtype: float64
```

```
df.groupby("Product")["Sales"].sum().nlargest(5)
```



Sales

Product	
Laptop	25350000
Refrigerator	15330000
Mobile Phone	7125000
Shoes	1235000
Headphones	708000

dtype: int64

```
df.pivot_table(values="Sales", index="City", columns="Product", aggfunc="sum")
```

Product	Headphones	Laptop	Mobile Phone	Refrigerator	Shoes	T-Shirt
City						
Bangalore	194000	8000000	2235000	4470000	360000	76800
Delhi	160000	5500000	2160000	3420000	322500	103800
Mumbai	148000	6300000	1470000	3900000	272500	86400
Pune	206000	5550000	1260000	3540000	280000	44400

```
df["Month"].value_counts()
```

count	
Month	
November	60
May	53
September	45
February	42
October	41
June	40
March	40
December	39
July	39
April	35
January	34
August	32

dtype: int64

```
df.groupby("OrderID")["Product"].apply(lambda x: ','.join(x))
```



Product	
OrderID	
1000	Laptop
1001	Refrigerator
1002	T-Shirt
1003	Mobile Phone

```
df.groupby("Product")["Quantity"].mean()
```

Quantity	
Product	
Headphones	5.283582
Laptop	5.827586
Refrigerator	5.873563
Mobile Phone	5.654762
Shoes	6.024390
T-Shirt	5.580645

dtype: object

dtype: float64

```
df.groupby("Product")["Price"].max().idxmax()
```

'Laptop'

```
df.groupby("Product")["Price"].min().idxmin()
```

'T-Shirt'

```
df.groupby("City")["Sales"].mean().idxmax()
```

'Bangalore'

```
df["Month"].value_counts().idxmax()
```

'November'

```
df["CustomerID"].value_counts().idxmax()
```

'C044'

```
df["Category"].value_counts(normalize=True)
```

proportion	
Category	
Electronics	0.476
Fashion	0.350
Appliances	0.174

dtype: float64

```
df.groupby("City")["Sales"].sum() / df["Sales"].sum() * 100
```

Sales	
City	
Bangalore	30.635205
Delhi	23.304914